

VAIBHAV LOHAR

Full-Stack Developer | AI/GenAI Integration | MCA Candidate

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PROFESSIONAL SUMMARY

Full-Stack Developer and MCA candidate (9.30 CGPA) with a proven track record of shipping production-ready systems — from an AI-powered e-commerce platform (ShopEase) to a GenAI portfolio agent built on a custom RAG pipeline with Supabase pgvector and Google Gemini embeddings. Skilled in end-to-end full-stack architecture using React.js, Node.js, Express.js, and PostgreSQL. Active problem solver with 670+ LeetCode problems solved and 1,600+ GitHub contributions in the last year. Passionate about integrating Generative AI into real-world products to automate workflows and deliver measurable impact.

TECHNICAL SKILLS

Frontend	HTML5, CSS3, JavaScript (ES6+), TypeScript, React.js, Vite, TailwindCSS, Framer Motion
Backend	Node.js, Express.js, RESTful API Design, JWT Authentication, bcrypt, Resend, Nodemailer
Databases	PostgreSQL, Supabase (pgvector), MongoDB
AI / GenAI	OpenAI API, Google Gemini API, RAG Architecture, Vector Search (pgvector), Prompt Engineering, LLM Integration
Tools	Git, GitHub, VS Code, Vercel, Render, Cloudinary, Postman, CI/CD
Core CS	Data Structures & Algorithms (DSA), System Design, Object-Oriented Programming (OOP)

PROJECTS

ShopEase – AI-Powered E-Commerce Platform [Live Demo](#) | [Admin Panel](#)

React.js · Node.js · Express.js · PostgreSQL · OpenAI API · Cloudinary · JWT · Vercel · Render

- **Architected** a monorepo with 3 decoupled applications — Customer Frontend, Admin Panel, and RESTful Backend — deployed independently on Vercel and Render with JWT dual-role authentication and bcrypt password hashing.
- **Built** a feature-rich Admin Dashboard (CMS) with modules for Product & Variant Management, Category Management, Coupon Engine (flat/percentage discounts with expiry), Audience Segmentation, and Design Gallery.
- **Engineered** a multi-variant inventory system supporting size, color, sub-size, variety, and custom SKU fields with real-time duplicate SKU validation and sub-200ms PostgreSQL query response times across 1,000+ SKUs.
- **Integrated** OpenAI API for automated product description generation (saving 50+ hours of manual work) and Cloudinary for optimized storage and delivery of product images and videos.
- **Delivered** end-to-end customer shopping flow — Catalog, Cart (React Context API), Wishlist, Checkout, Payment, and Shipping — with skeleton loaders and Resend transactional email notifications.
- **Achieved** 95+ Lighthouse Performance Score through SSR, lazy loading, and Cloudinary image optimization, resulting in 25% faster page loads.

GenAI-Powered Portfolio & Developer Agent [Live Link](#)

React.js · Vite · Google Gemini API · Supabase pgvector · RAG Architecture · Framer Motion · Vercel

- **Designed** a custom AI Agent powered by Google Gemini API with intent detection across 6 query categories, delivering real-time answers about projects, skills, and experience with zero hallucinations.
- **Implemented** a RAG (Retrieval-Augmented Generation) pipeline using Supabase pgvector for cosine similarity search and Gemini embeddings to ground every AI response in verified portfolio data.

- **Engineered** a structured knowledge base layer for intent-based retrieval, enabling context-aware responses without triggering full vector search on every query — optimizing latency and API costs.
- **Integrated** live GitHub & LeetCode APIs with graceful fallbacks to display real-time contribution graphs, repository stats, streak data, and problem-solving metrics in the portfolio UI.
- **Architected** a serverless API layer on Vercel (/api/stats.js) to proxy GitHub and LeetCode GraphQL queries, keeping API keys server-side and eliminating CORS issues.

EDUCATION

Master of Computer Applications (MCA) – Software Engineering

Aug 2024 – Jun 2026 (Expected)

R. C. Patel Institute of Technology / IMRD College, Shirpur

CGPA: 9.30 / 10

Bachelor of Computer Science (BCS)

Aug 2020 – Jun 2023

North Maharashtra University, Jalgaon

CGPA: 9.08 / 10

ACHIEVEMENTS

- **1st Place – Webtech Developers Pvt. Ltd. C++ Project Challenge (July 2025):** Secured 1st place among numerous participants for developing a 3D cargo-loading optimization desktop application using C++ and Qt6 with real-time bin-packing visualization and PDF export.
- **Top Academic Performer:** Consistently maintained a 9.30 CGPA in MCA and 9.08 CGPA in BCS (A+ Grade), ranking among the top students across both programs.
- **Active Open-Source Contributor:** 1,600+ GitHub contributions and 670+ LeetCode problems solved in the last year, demonstrating consistency and a strong problem-solving mindset.